

TP_Ansible

HIBAT-ALLAH JAMIL

Installation

Générer une clé SSH

```
[node1] (local) root@192.168.0.28 ~
$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
/root/.ssh/id_rsa already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:IJKtC2cbrfC9so6xnXYS/G5EEFA4ktu8qmKNEEL6F5w root@node1
The key's randomart image is:
+---[RSA 3072]-----+
|.o.                  |
|=.o                  |
|+=ooo..              |
```

```
[node1] (local) root@192.168.0.28 ~
$ ls -la /root/.ssh
total 12
drwxr-xr-x  1 root  root    38 Jan 26 14:59 .
drwx-----  1 root  root    18 Jan 26 14:59 ..
-rw-r--r--  1 root  root   102 Jan 26 14:59 authorized_keys
-rw-----  1 root  root  2635 May 19 08:29 id_rsa
-rw-r--r--  1 root  root   564 May 19 08:29 id_rsa.pub
[node1] (local) root@192.168.0.28 ~
```

Copier la clé dans les noeuds

```
[node1] (local) root@192.168.0.13 ~
$ ssh-copy-id -i /root/.ssh/id_rsa.pub root@192.168.0.11
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
expr: warning: '^ERROR: ': using '^' as the first character
of a basic regular expression is not portable; it is ignored
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new k
eys
root@192.168.0.11's password:
Number of key(s) added: 1
Now try logging into the machine, with:  "ssh 'root@192.168.0.11'"
and check to make sure that only the key(s) you wanted were added.
[node1] (local) root@192.168.0.13 ~
```

```
[node1] (local) root@192.168.0.13 ~
$ ssh-copy-id -i /root/.ssh/id_rsa.pub root@192.168.0.12
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub"
The authenticity of host '192.168.0.12 (192.168.0.12)' can't be established.
ED25519 key fingerprint is SHA256:1zFux1EgS8o8KQFDx1TmIJlcNQ1+9Lte6/ik72mG360.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: 192.168.0.11
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
expr: warning: '^ERROR: ': using '^' as the first character
of a basic regular expression is not portable; it is ignored
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new key(s)
root@192.168.0.12's password:
Number of key(s) added: 1
```

```
$ ssh root@192.168.0.12
#####
#                               #
#      WARNING!!!!              #
# This is a sandbox environment. Using personal credentials #
# is HIGHLY! discouraged. Any consequences of doing so are #
# completely the user's responsibilities.                     #
#                                                             #
# The PWD team.                                                #
#####
-sh: disown: not found
[node2] (local) root@192.168.0.12 ~
$
[1]+  Done(1) (while !docker info 1>&/dev/null; do sleep 1; done; docker network create -d bri
dge docker_gwbridge 1>&/dev/null)
[node2] (local) root@192.168.0.12 ~
$ exit
Connection to 192.168.0.12 closed.
[node1] (local) root@192.168.0.13 ~
```

Installer Ansible

```
[node1] (local) root@192.168.0.13 ~
$ apk update
fetch https://dl-cdn.alpinelinux.org/alpine/v3.20/main/x86_64/APKINDEX.tar.gz
fetch https://dl-cdn.alpinelinux.org/alpine/v3.20/community/x86_64/APKINDEX.tar.gz
v3.20.6-151-g3271406ca0d [https://dl-cdn.alpinelinux.org/alpine/v3.20/main]
v3.20.6-154-gcccb156823d [https://dl-cdn.alpinelinux.org/alpine/v3.20/community]
OK: 24215 distinct packages available
[node1] (local) root@192.168.0.13 ~
$ apk add ansible
(1/25) Installing py3-cparser (2.22-r1)
(2/25) Installing py3-cparser-pyc (2.22-r1)
(3/25) Installing py3-cffi (1.16.0-r1)
(4/25) Installing py3-cffi-pyc (1.16.0-r1)
(5/25) Installing py3-cryptography (42.0.7-r0)

[node1] (local) root@192.168.0.13 ~
$ ansible --version
ansible [core 2.17.0]
  config file = None
  configured module search path = ['/root/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3.12/site-packages/ansible
  ansible collection location = /root/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.6 (main, Sep 11 2024, 08:48:26) [GCC 13.2.1 20240309] (/usr/bin/python3)
  jinja version = 3.1.6
  libyaml = True
[node1] (local) root@192.168.0.13 ~
$
```

Premiers pas

```
[node1] (local) root@192.168.0.13 ~
$ ansible all -i 192.168.0.11, -m ping
[WARNING]: Platform linux on host 192.168.0.11 is using the discovered Python interpreter at
/usr/bin/python3.12, but future installation of another Python interpreter could change the meaning of that
path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more
information.
192.168.0.11 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.12"
  },
  "changed": false,
  "ping": "pong"
}
[node1] (local) root@192.168.0.13 ~
$
```

```
[node1] (local) root@192.168.0.13 ~
$ ansible all -i 192.168.0.12, -m ping
[WARNING]: Platform linux on host 192.168.0.12 is using the discovered Python interpreter at
/usr/bin/python3.12, but future installation of another Python interpreter could change the meaning of that
path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more
information.
192.168.0.12 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.12"
  },
  "changed": false,
  "ping": "pong"
}
[node1] (local) root@192.168.0.13 ~
$
```

```
$ ansible all -i 192.168.0.12, -m user -a "name=hiba state=present"
[WARNING]: Platform linux on host 192.168.0.12 is using the discovered Python interpreter at
/usr/bin/python3.12, but future installation of another Python interpreter could change the meaning of that
path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more
information.
192.168.0.12 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.12"
  },
  "changed": true,
  "comment": "Linux User,,,",
  "create_home": true,
  "group": 1000,
  "home": "/home/hiba",
  "name": "hiba",
  "shell": "/bin/sh",
  "state": "present",
  "system": false,
}
```

```
[node1] (local) root@192.168.0.13 ~
$ ansible all -i 192.168.0.12, -m user -a "name=hiba state=absent"
[WARNING]: Platform linux on host 192.168.0.12 is using the discovered Python interpreter at
/usr/bin/python3.12, but future installation of another Python interpreter could change the meaning of that
path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more
information.
192.168.0.12 | CHANGED => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.12"
  },
  "changed": true,
  "force": false,
  "name": "hiba",
  "remove": false,
  "state": "absent",
  "stderr": "deluser: can't find hiba in /etc/group\n",
  "stderr_lines": [

```

```
[node1] (local) root@192.168.0.13 ~
$ ansible all -i 192.168.0.12, -m user -a "name=hiba state=absent"
[WARNING]: Platform linux on host 192.168.0.12 is using the discovered Python interpreter at
/usr/bin/python3.12, but future installation of another Python interpreter could change the meaning of that
path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more
information.
192.168.0.12 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.12"
  },
  "changed": false,
  "name": "hiba",
  "state": "absent"
}
[node1] (local) root@192.168.0.13 ~
$
```

Playbook

```
[node1] (local) root@192.168.0.13 ~
$ apk add mc
(1/6) Installing glib (2.80.5-r0)
(2/6) Installing glib-bash-completion (2.80.5-r0)
(3/6) Installing gpm-libs (1.20.7-r4)
(4/6) Installing slang (2.3.3-r3)
(5/6) Installing libssh2 (1.11.0-r2)
(6/6) Installing mc (4.8.31-r0)
Executing busybox-1.36.1-r29.trigger
```



```
[node1] (local) root@192.168.0.13 ~/folder1
$ mcedit inventory

[node1] (local) root@192.168.0.13 ~/folder1
$ cat inventory
[nodes]
192.168.0.11
192.168.0.12 [node1] (local) root@192.168.0.13 ~/folder1
```

```
inventory  [-M--] 12 L: [ 1+ 2 3/ 3] *(33 / 33b) <EOF> [*][X]
[nodes]
192.168.0.11
192.168.0.12
1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

```
ping.yml  [-M--] 4 L: [ 1+ 5 6/ 8] *(74 / 95b) 0115 [*][X]
-----
- name: ping sur groupe de nodes
hosts: nodes
remote_user: root
.
tasks:
- name: ping
ping:
```

```
$ mcedit ping.yml

[node1] (local) root@192.168.0.13 ~/folder1
$ tree
.
├── inventory
└── ping.yml

1 directory, 2 files
[node1] (local) root@192.168.0.13 ~/folder1
$
```

```
[node1] (local) root@192.168.0.13 ~/folder1
$ ansible-playbook -i inventory ping.yml --syntax-check

playbook: ping.yml
[node1] (local) root@192.168.0.13 ~/folder1
```

```
[node1] (local) root@192.168.0.13 ~/folder1
ansible-playbook -i inventory ping.yml

PLAY [ping sur groupe de nodes] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host 192.168.0.11 is using the discovered Python interpreter at
/usr/bin/python3.12, but future installation of another Python interpreter could change the meaning of that
path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more
information.
ok: [192.168.0.11]
[WARNING]: Platform linux on host 192.168.0.12 is using the discovered Python interpreter at
/usr/bin/python3.12, but future installation of another Python interpreter could change the meaning of that
path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more
information.
ok: [192.168.0.12]
```

Installation d'un serveur web

1. Installer un serveur web apache2 (modules apk et shell)

```
install.yml  [-M--]  3 L:[ 1+ 0  1/ 24] +(3  / 500b) 0010 0x00A
---
- name: installation serveur web
  hosts: nodes
  remote_user: root
  vars:
    user: node
  tasks:
    - name: mettre à jour le cache des paquets apk
      apk:
        update_cache: yes
      tags: update
    - name: installation Apache et php
      apk:
        name:
          - apache2
          - php83
```

```
tree
---
-- hosts
-- install.yml

. directory, 2 files
[node1] (local) root@192.168.0.18 ~/serveurweb
```

```
[node1] (local) root@192.168.0.18 ~/serveurweb
$ ansible-playbook -i hosts install.yml --syntax-check

playbook: install.yml
[node1] (local) root@192.168.0.18 ~/serveurweb
$ ansible-playbook -i hosts install.yml

PLAY [installation serveur web] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host 192.168.0.17 is using the discovered Python interpreter at
/usr/bin/python3.12, but future installation of another Python interpreter could change the meaning of that
path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more
information.
ok: [192.168.0.17]
```

2. Ajouter un utilisateur Linux (module user)

```
[node1] (local) root@192.168.0.18 ~/serveurweb
$ ssh 192.168.0.17 "cat /etc/passwd | grep node"
node:x:1000:1000:Linux User,,,:/home/node:/bin/false
[node1] (local) root@192.168.0.18 ~/serveurweb
$ ssh 192.168.0.16 "cat /etc/passwd | grep node"
node:x:1000:1000:Linux User,,,:/home/node:/bin/false
[node1] (local) root@192.168.0.18 ~/serveurweb
$
```

3. Copier un fichier sur les noeuds (module file et copy)

```
hosts      [----] 34 L: [ 1+ 2  3/ 3] *(77 / 77b) <EOF>
[nodes]
192.168.0.17 domain="192.168.0.17"
192.168.0.16 domain="192.168.0.16"
```

```
install.yml  [-M--] 17 L: [ 23+15  38/ 38] *(837 / 837b) <EOF[*][X]
  password: !
  shell: /bin/false
- name: creer dossier user
  file:
    path: /var/www/[[ domain ]]
    state: directory
    owner: "[[ user ]]"
    group: "[[ user ]]"
    mode: 0755
- name: phpinfo
  copy:
    src: files/index.php
    dest: /var/www/[[ domain ]]
    owner: "[[ user ]]"
    group: "[[ user ]]"
    mode: 0644
1Help 2Save 3Mark 4Re~ac 5Copy 6Move 7Se~ch 8De~te 9Pu~Dn 10Quit
```

```
index.php  [-M--] 1 L: [ 1+ 2  3/ 3] *(18 / 18b) 0062 0x03E
<?php
phpinfo();
?>
```

```
[node1] (local) root@192.168.0.18 ~/serveurweb
$ tree
.
├── files
│   └── index.php
├── hosts
└── install.yml

2 directories, 3 files
```

